

A CONDITIONAL FINITENESS THEOREM FOR PARADOXICAL COLLATZ SEQUENCES WITH A BOUNDED NUMBER OF CIRCUITS

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ABSTRACT. For the Collatz function T , Rozier and Terracol call a finite sequence $\Omega_j(n) = (n, T(n), \dots, T^j(n))$ with $n > 2$ *paradoxical* when its coefficient $C_j(n) = 3^q/2^j$ is less than 1 and yet $T^j(n) \geq n$. Fix $R \geq 1$ and let $\mathcal{P}_R$ be the set of paradoxical sequences whose parity vector is a word of R circuits $1^{a_1}0^{e_1} \dots 1^{a_R}0^{e_R}$. We prove: if, for some $\beta \geq 0$, every member of $\mathcal{P}_R$ satisfies the excursion bound $M_j(n) \leq m_j(n)^\beta$ — where $M_j(n)$ and $m_j(n)$ denote the greatest and smallest terms of $\Omega_j(n)$ — then $\mathcal{P}_R$ is finite. The two remaining links are unconditional: a run-length pigeonhole argument shows $2^{j/(2R)} - 1 \leq M_j(n)$ on the slice, and an application of Rhin’s effective irrationality measure for $\log 2/\log 3$ to the denominator of Corollary 4.4 of Rozier–Terracol yields the polynomial bound $m_j(n) \leq \frac{1}{3}j^{14.3}$; an exponential–polynomial collision then caps the length j . The only transcendence input is Rhin’s bound; all results are machine-verified in Lean 4.

1. INTRODUCTION

Among the necessary conditions that Rozier and Terracol [2] derive for a Collatz sequence to be paradoxical, two have a quantitative character: the ones-ratio of the parity vector is pinned close to $\log 2/\log 3$ [2, Cor. 4.3], and the harmonic mean of the odd terms is bounded polynomially in the length [2, Cor. 4.4]. In §6 of [2] these are combined, through the effective irrationality measure of Rhin for linear forms in $\log 2$ and $\log 3$ [1] and conditionally on a heuristic delay bound ([2, Conj. 6.2]), into an argument that only finitely many paradoxical sequences should exist.

This note isolates the part of that programme that can be carried out rigorously for sequences with a *bounded number of circuits*. Writing the parity vector of a paradoxical sequence as a word of R circuits $1^{a_1}0^{e_1} \dots 1^{a_R}0^{e_R}$, we assemble three links:

- (1) a *run-length pigeonhole* (unconditional): some circuit carries a run of one parity of length at least $j/(2R)$, and a long run of either parity forces a large term, so the greatest term of the window satisfies $2^{j/(2R)} - 1 \leq M_j(n)$ (§§3–4);

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- (2) a *polynomial bound on the smallest term* (unconditional modulo Rhin's theorem): $m_j(n) \leq \frac{1}{3}j^{14.3}$ (§5);
- (3) an *excursion hypothesis* (the sole conjectural link): every member of the slice satisfies $M_j(n) \leq m_j(n)^\beta$ for a fixed $\beta \geq 0$ — a windowed form of the bound $M_T(n) \leq n^\beta$ of [2, eq. (16)], itself a consequence of [2, Conj. 6.2].

Chaining (1)–(3) traps an exponential in j below a polynomial in j , which bounds j ; since a paradoxical sequence of length j has first term at most $2^j 3^j$, the slice is finite (Theorem 6.3). A bound uniform over *all* R cannot be expected from this route: it is equivalent to the finiteness of the whole paradoxical set, hence implies the Collatz conjecture (Remark 6.5).

2. NOTATION AND RECALLED RESULTS

We follow the notation of [2]. Let T be the Collatz function

$$T(n) = \begin{cases} \frac{3n+1}{2}, & n \text{ odd}, \\ \frac{n}{2}, & n \text{ even}, \end{cases}$$

on the positive integers, $\Omega_j(n) = (T^k(n))_{k=0}^j$ the sequence of iterates of length $j+1$ [2, Def. 4.1], $q_j(n)$ (or q) the number of odd terms among $n, T(n), \dots, T^{j-1}(n)$, the *coefficient* $C_j(n) = 3^q/2^j$ [2, eq. (2)], and $V_j(n) \in \{0, 1\}^j$ the *parity vector* of n [2, Def. 2.1], written as a binary word with repetitions abbreviated by superscripts.

Definition 2.1 ([2, Def. 1.1]). For positive integers j and n with $n > 2$, the sequence $\Omega_j(n)$ is *paradoxical* if

- (i) $C_j(n) < 1$, and
- (ii) $T^j(n) \geq n$.

We write $\mathcal{P} = \{(j, n) : \Omega_j(n) \text{ paradoxical}, n > 2\}$ for the *paradoxical set*. Several intermediate statements below assume only conditions (i)–(ii), with $n \geq 1$; we then say that $\Omega_j(n)$ satisfies (i)–(ii).

Any $\Omega_j(n)$ satisfying (i)–(ii) with $n \geq 1$ has $j \geq 2$: for $j = 0$ one has $C_0(n) = 1$, and for $j = 1$ either n is odd and $C_1(n) = 3/2$, or n is even and $T(n) = n/2 < n$.

As in the companion note on canonical circuit decompositions, a *circuit* is a pair (a, e) of nonnegative integers, spelling the word $1^a 0^e$; the terminology extends the classical usage, in which a circuit is a (near-)cyclic sequence with a single ascent–descent phase, i.e. with parity vector one block $1^a 0^{j-a}$ [2, App. A].¹ A finite list $S = ((a_1, e_1), \dots, (a_R, e_R))$ of circuits spells the word $w(S) = 1^{a_1} 0^{e_1} \dots 1^{a_R} 0^{e_R}$ of length $|S| := \sum_i (a_i + e_i)$.

¹We are not aware of standard terminology for the individual blocks of a general parity vector and adopt *circuit* for them.

Definition 2.2. For $R \geq 0$, the R -circuit slice $\mathcal{P}_R \subseteq \mathcal{P}$ is the set of pairs $(j, n) \in \mathcal{P}$ such that $V_j(n) = w(S)$ for some list S of R circuits with $|S| = j$. (Blocks may be empty, so the slices overlap; only their union matters below.)

Following [2, Def. 5.2], the *delay* $d_T(n)$ is the least j with $T^j(n) = 1$ (or $+\infty$), and the *maximum excursion* $M_T(n)$ is the greatest term of the infinite sequence $(T^k(n))_{k \geq 0}$ (or $+\infty$). We shall use their windowed counterparts

$$M_j(n) = \max_{0 \leq k \leq j} T^k(n), \quad m_j(n) = \min_{0 \leq k \leq j} T^k(n),$$

the greatest and smallest terms of $\Omega_j(n)$; thus $M_T(n) = \sup_j M_j(n)$. Heuristic and numerical evidence suggests that the delay satisfies $d_T(n) \leq \alpha \log n$ for an absolute constant α [2, Conj. 6.2], from which one derives a polynomial excursion bound $M_T(n) \leq n^\beta$ [2, eq. (16)].

The transcendence input is the effective irrationality measure of Rhin, as stated in [2, Prop. 6.3].

Proposition 2.3 (Rhin [1]; see [2, Prop. 6.3]). *If u_0, u_1, u_2 are rational integers such that $\max(|u_1|, |u_2|) \geq 2$, then the form $\Lambda = u_0 + u_1 \log 2 + u_2 \log 3$ verifies*

$$|\Lambda| \geq \max(|u_1|, |u_2|)^{-13.3}.$$

Finally we recall two facts from the companion formal development. The first is the smallest-term form of [2, Cor. 4.4]: the original bounds the harmonic mean h of the odd terms of $\Omega_{j-1}(n)$, and [2] note in §4 that one often uses the weakened form in which h is replaced by the smallest term. Write, as in [2, eq. (12)],

$$H(j) = \frac{1}{2^{r(j)} - 3}, \quad r(j) = \frac{j}{\lfloor j \log 2 / \log 3 \rfloor}.$$

Proposition 2.4 (smallest-term form of [2, Cor. 4.4]). *Let $j \geq 2$, $n \geq 1$, and let $m \geq 1$ be a common lower bound for the odd terms among $n, T(n), \dots, T^{j-1}(n)$. If $\Omega_j(n)$ satisfies (i)–(ii), then $m \leq H(j)$.*

Lemma 2.5. *If $\Omega_j(n)$ is paradoxical with $n > 2$, then $n \leq 2^j 3^j$.*

3. LONG RUNS FORCE LARGE TERMS

The arithmetic core of link (1) is that a run of consecutive steps of one parity forces a large orbit term: e successive halvings require $2^e \mid T^p(n)$, while a successive odd steps require $2^a \mid T^p(n) + 1$.

Lemma 3.1. *Let $n, e, p \geq 0$. If $T^{p+i}(n)$ is even for all $i < e$, then $2^e \mid T^p(n)$.*

Proof sketch. Induction on e : each even step halves the term, so $T^p(n) = 2T^{p+1}(n)$ and the inductive divisibility at $p+1$ doubles. $\square$

Lemma 3.2. *Let $n, a, p \geq 0$. If $T^{p+i}(n)$ is odd for all $i < a$, then $2^a \mid T^p(n) + 1$.*

Proof sketch. Induction on a . An odd step $x \mapsto (3x + 1)/2$ sends $x + 1$ to $\frac{3}{2}(x + 1)$: with $x = T^p(n)$ one checks $T^{p+1}(n) + 1 = 3(\lfloor x/2 \rfloor + 1)$, so the inductive divisibility $2^{a-1} \mid T^{p+1}(n) + 1$ passes to $2^{a-1} \mid \lfloor x/2 \rfloor + 1$ (as 2^{a-1} is coprime to 3) and doubles to $2^a \mid x + 1$. $\square$

In particular a run of e even steps forces $T^p(n) \geq 2^e$ (for $n \geq 1$), a run of a odd steps forces $T^p(n) \geq 2^a - 1$, and the maximum excursion of Definition 5.2 of [2] satisfies $M_T(n) \geq 2^e$, respectively $M_T(n) \geq 2^a - 1$.

4. THE RUN-LENGTH PIGEONHOLE

Throughout this section S is a nonempty list of R circuits with total length $j = |S|$, and $n \geq 1$ satisfies $V_j(n) = w(S)$. Paradoxicality is not needed here. The i -th circuit (a_i, e_i) of S contributes to $V_j(n)$ a run of a_i odd steps followed by a run of e_i even steps, located at position $\sum_{i' < i} (a_{i'} + e_{i'})$ of the orbit.

Proposition 4.1. *There are $L, p \geq 0$ with*

$$j \leq 2RL, \quad p \leq j, \quad 2^L - 1 \leq T^p(n).$$

Proof sketch. By the pigeonhole principle some circuit (a_i, e_i) has $a_i + e_i \geq j/R$; let $L = \max(a_i, e_i) \geq j/(2R)$. If the even run is the larger, Lemma 3.1 at its starting position p gives $2^L \mid T^p(n)$, so $T^p(n) \geq 2^L$; if the odd run is the larger, Lemma 3.2 gives $2^L \mid T^p(n) + 1$, so $T^p(n) \geq 2^L - 1$. In both cases p lies inside the window, $p \leq j$. $\square$

Proposition 4.2. *With S , $j = |S|$, R , n as above,*

$$2^{j/(2R)} - 1 \leq M_j(n).$$

Proof sketch. Take L, p from Proposition 4.1. Then $2^{j/(2R)} \leq 2^L$ since $j \leq 2RL$, and $2^L - 1 \leq T^p(n) \leq M_j(n)$ since $p \leq j$. $\square$

5. THE RHIN POLYNOMIAL BOUND ON THE SMALLEST TERM

Link (2) bounds $H(j)$, and with it the smallest odd term of a sequence satisfying (i)–(ii), polynomially in j .

Proposition 5.1. *For $j \geq 2$,*

$$H(j) \leq \frac{j^{14.3}}{3}.$$

Proof sketch. Put $Q = \lfloor j \log 2 / \log 3 \rfloor$, so $1 \leq Q < j$ (using $j \geq 2$ and $\log 2 / \log 3 < 1$), and $\Lambda_0 = j \log 2 - Q \log 3 \geq 0$ by the floor. Applying Proposition 2.3 with $(u_0, u_1, u_2) = (0, j, -Q)$ — of height $\max(j, Q) = j \geq 2$ — gives $\Lambda_0 \geq j^{-13.3}$. On the other hand $2^{r(j)} = 3e^{\Lambda_0/Q}$, so by $e^x \geq 1 + x$,

$$2^{r(j)} - 3 \geq \frac{3\Lambda_0}{Q}, \quad \text{whence} \quad H(j) \leq \frac{Q}{3\Lambda_0} \leq \frac{j}{3} \cdot j^{13.3} = \frac{j^{14.3}}{3}. \quad \square$$

Proposition 5.2. *Let j, n, m be integers with $n \geq 1$, $m \geq 1$, such that every odd term among $n, T(n), \dots, T^{j-1}(n)$ is at least m . If $\Omega_j(n)$ satisfies (i)–(ii), then*

$$m \leq \frac{1}{3}j^{14.3}.$$

Proof sketch. $j \geq 2$ as noted in §2; chain Proposition 2.4 with Proposition 5.1. $\square$

Remark 5.3. The bound of Proposition 5.1 does not appear in [2]: in §6 they apply Rhin’s estimate to the form $\Lambda = j \log 2 - q \log 3$ attached to the sequence itself. Since Proposition 2.3 holds for arbitrary integer coefficients, it may instead be fed the denominator of [2, eq. (12)], whose exponent $r(j)$ involves the floor $\lfloor j \log 2 / \log 3 \rfloor$ rather than the odd-step count q of any particular sequence; this makes the bound on $H(j)$ — and hence on the smallest odd term — a function of the length alone. The exponent is $14.3 = 13.3 + 1$.

6. THE COLLISION ENGINE AND THE MAIN THEOREM

The analytic engine is elementary: an exponential trapped below a polynomial bounds the argument.

Lemma 6.1. *Let $a > 0$ and $K, Q \geq 0$. There is a $J \in \mathbb{N}$ such that every $j \in \mathbb{N}$ with $e^{aj} - 1 \leq Kj^Q$ satisfies $j \leq J$.*

Proof sketch. $t^Q e^{-at} \rightarrow 0$ as $t \rightarrow \infty$, so beyond some M one has $t^Q e^{-at} < 1/(K+1)$; take $J = \lceil M \rceil$. For $j > J$ this yields $(K+1)j^Q < e^{aj}$, while the assumed inequality (together with $j^Q \geq 1$) gives $e^{aj} \leq Kj^Q + 1 \leq (K+1)j^Q$, a contradiction. $\square$

We state the finiteness engine for an abstract class of sequences, with the excursion and smallest-term functions as parameters; it will be instantiated on the R -circuit slice.

Theorem 6.2. *Let $R \geq 1$ be an integer, let $\beta, P_w \geq 0$ and $C > 0$ be real constants, and let P be a set of pairs (j, n) such that every $(j, n) \in P$ satisfies (i)–(ii) with $n > 2$. Let $\text{Exc}, m: P \rightarrow \mathbb{R}$ be functions such that, for all $(j, n) \in P$,*

- (1) $2^{j/(2R)} - 1 \leq \text{Exc}(j, n)$,
- (2) $1 \leq m(j, n) \leq C j^{P_w}$, and
- (3) $\text{Exc}(j, n) \leq m(j, n)^\beta$.

Then P is finite.

Proof sketch. Write $2^{j/(2R)} = e^{aj}$ with $a = \log 2 / (2R) > 0$. Chaining (1), (3), (2) gives

$$e^{aj} - 1 \leq \text{Exc}(j, n) \leq m(j, n)^\beta \leq (C j^{P_w})^\beta = C^\beta j^{P_w \beta},$$

so Lemma 6.1 bounds $j \leq J$ on P . By Lemma 2.5, $n \leq 2^j 3^j \leq 2^J 3^J$, and P lies in a finite rectangle. $\square$

Instantiating on the slice yields the main theorem. Its sole hypothesis is the excursion bound; the other two links are Proposition 4.2 and Proposition 5.2.

Theorem 6.3. *Let $R \geq 1$ be an integer and $\beta \geq 0$ a real constant. Assume that every $(j, n) \in \mathcal{P}_R$ satisfies the excursion bound*

$$M_j(n) \leq m_j(n)^\beta.$$

Then $\mathcal{P}_R$ is finite.

Proof sketch. Apply Theorem 6.2 with $P = \mathcal{P}_R$, $\text{Exc} = M_j(n)$, $m = m_j(n)$, $C = \frac{1}{3}$ and $P_w = 14.3$. Hypothesis (1) is Proposition 4.2 applied to the witnessing circuit list of Definition 2.2. For (2): all terms of $\Omega_j(n)$ are at least 1, so $1 \leq m_j(n)$; and since the smallest term of the window is a common lower bound for its odd terms, Proposition 5.2 gives $m_j(n) \leq \frac{1}{3}j^{14.3}$. Hypothesis (3) is the assumption. $\square$

Remark 6.4. The hypothesis of Theorem 6.3 is a windowed form of the polynomial excursion bound $M_T(n) \leq n^\beta$ of [2, eq. (16)] (a consequence of the delay bound [2, Conj. 6.2]), with the starting value replaced by the smallest term of the window. Numerical experiments on the computed portion of the slice are consistent with a value $\beta \in [1.58, 2.06]$.

Remark 6.5. The per- R statement is the honest endpoint of this route. In the companion note on canonical circuit decompositions it is shown that a length bound valid on $\mathcal{P}_R$ uniformly in R is *equivalent* to the finiteness of the whole paradoxical set, which by [2, Cor. 3.3] implies the Collatz conjecture; and assuming the excursion hypothesis uniformly in R does not produce such a bound through the present argument, since for $R \approx j/2$ the pigeonhole exponent $j/(2R)$ degenerates to a constant.

APPENDIX A. FORMAL STATEMENTS

The results of this note are machine-verified in Lean 4. The assembly lives in `BakerReduction.lean`, in directory `paradoxical/` of the author's formal development, together with the files it builds upon: `Defs`, `RCircuitSlice`, `ExcursionBounds`, `RunPigeonhole`, `PigeonholeFiniteness`, `RhinPolyBound`.

All proofs are complete; the sole extra-logical axiom in the development is Rhin's theorem (Proposition 2.3), recorded as a cited axiom on the authority of [1]. Its phrasing was checked against the verbatim restatement [2, Prop. 6.3]; the original [1] was not available to this write-up.

Declarations live in the Lean namespace `Paradoxical` unless prefixed otherwise.

Environment	Lean declaration	status
Def. 2.1	<code>RT.IsParadoxical</code>	definition (companion)
Def. 2.2	<code>RCircuitSlice</code>	definition
$M_j(n), m_j(n)$	<code>excWin, minWin</code>	definitions
$H(j), r(j)$	<code>RT.H, RT.rExp</code>	definitions (companion)
Prop. 2.3	<code>Rhin.logForm_lower_bound</code>	cited axiom [Rhi87]
Prop. 2.4	<code>RT.m_le_H</code>	proved (companion)
Lem. 2.5	<code>RT.paradoxical.start_le</code>	proved (companion)
Lem. 3.1	<code>two_pow_dvd_of_even_run</code>	proved
Lem. 3.2	<code>two_pow_dvd_succ_of_odd_run</code>	proved
Prop. 4.1	<code>shape_excursion_bound</code>	proved
Prop. 4.2	<code>pigeonhole_hyp</code>	proved
Prop. 5.1	<code>H.le.rpow</code>	proved, modulo Rhin
Prop. 5.2	<code>least_odd_poly_bound</code>	proved, modulo Rhin
Lem. 6.1	<code>exp_poly_collision</code>	proved
Thm. 6.2	<code>finite_of_pigeonhole</code>	proved
Thm. 6.3	<code>finite_RCircuitSlice</code>	proved, modulo Rhin

REFERENCES

- [1] G. Rhin, *Approximants de Padé et mesures effectives d'irrationalité*, Séminaire de Théorie des Nombres, Paris 1985–86, 155–164, Progr. Math. 71, Birkhäuser Boston, Boston, MA, 1987.
- [2] O. Rozier and C. Terracol, *Paradoxical behavior in Collatz sequences*, arXiv:2502.00948 (2025).